

Software Requirement Specification (SRS)

1. Introduction

The **Men's Shopping Mart – E-Commerce Database Management System** is a centralized database designed to manage the daily activities of an online men's fashion store. The system stores customer details, product information, categories, suppliers, inventory, orders, payments, shipments, and customer reviews. It helps maintain accurate records, reduces data duplication, and improves business efficiency.

2. Purpose

The purpose of this Software Requirement Specification (SRS) is to define the software requirements for the **Men's Shopping Mart – E-Commerce Database Management System**. It serves as a guideline for database design, ER modeling, SQL implementation, testing, and future maintenance.

3. Scope

The system provides the following features:

- Customer registration and profile management.
- Product and category management.
- Supplier information management.
- Inventory and stock management.
- Customer order processing.
- Order details management.
- Payment recording and status tracking.
- Shipment and delivery management.
- Customer review management.
- Business report generation.

4. Overall System Description

The system maintains all business data in a centralized relational database using the following entities:

- Customer
- Category
- Product
- Supplier
- Order
- Order Details
- Payment
- Shipment

- Review

Authorized users can add, update, delete, and retrieve records while maintaining data consistency and integrity.

5. Functional Requirements

5.1 Customer Management

- Register new customers.
- Update customer information.
- Maintain customer records.

5.2 Category Management

- Create product categories.
- Assign products to categories.

5.3 Product Management

- Add and update product details.
- Manage product price and stock.

5.4 Supplier Management

- Store supplier information.
- Maintain supplier contact details.

5.5 Inventory Management

- Track stock availability.
- Update inventory after orders.

5.6 Order Management

- Create customer orders.
- Maintain order status.
- Store order details.

5.7 Payment Management

- Record payment information.
- Track payment status.

5.8 Shipment Management

- Maintain shipment details.
- Update delivery status.

5.9 Review Management

- Store product ratings.
- Record customer feedback.

5.10 Report Generation

Generate reports for:

- Customers
 - Products
 - Orders
 - Payments
 - Shipments
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6. Non-Functional Requirements

Security

- Only authorized users can access the database.
- Customer information must remain confidential.

Performance

- The system should process records quickly.

Reliability

- Data should remain accurate and consistent.

Availability

- The database should be available during business operations.

Data Integrity

- Relationships between all entities must be maintained.

Scalability

- The system should support future expansion.

Maintainability

- Database modifications should be simple.

Backup & Recovery

- Regular backups should prevent data loss.

Usability

- The system should be easy to use.

Compatibility

- The database should support standard DBMS platforms.
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7. Assumptions

- Customers provide valid registration details.
 - Products are categorized before listing.
 - Stock is updated after every purchase.
 - Every order belongs to one customer.
 - Payment information is entered correctly.
 - Shipment details are updated after dispatch.
 - Reviews are submitted by registered customers only.
 - Authorized users follow system guidelines.
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8. Constraints

- Only the predefined entities and attributes are used.
 - Entity names and attributes cannot be modified.
 - The project focuses only on database management.
 - Online payment gateway integration is not included.
 - Internet access is required.
 - Report accuracy depends on stored data.
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9. Conclusion

The **Men's Shopping Mart – E-Commerce Database Management System** provides a secure and efficient solution for managing online men's fashion store operations. The system supports customer management, inventory tracking, order processing, payments, shipments, and reviews through a centralized database. This SRS serves as the foundation for successful

database design, implementation, testing, and future enhancements while ensuring reliable and organized data management.